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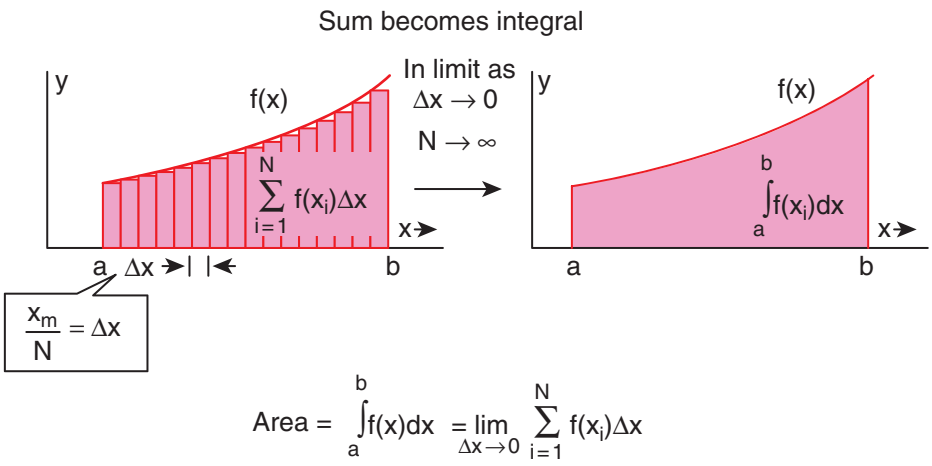


Figure 2.31 Relation between summation and integration

Examples

In physics the work done by a force F(x) on an object to move it from point a to point b in one dimension is given by

$$W = \int_a^b F(x)\,dx$$

(No scalar products is required here, since motion here is in one dimension)

1) The work done is the area under the force displacement graph as shown in Figure 2.32

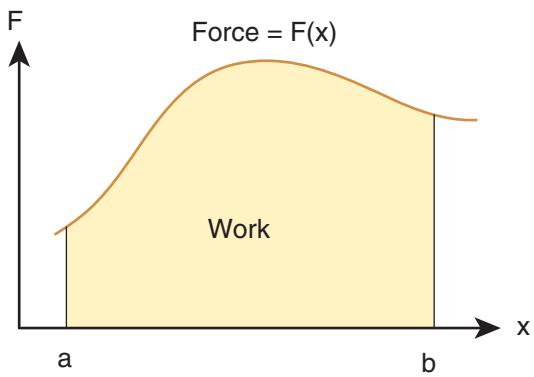


Figure 2.32 Work done by the force

2) The impulse given by the force in an interval of time is calculated between the interval from time t = 0 to time t = t_1 as

$$\text{Impulse } I = \int_0^{t_1} F\,dt$$

The impulse is the area under the force function F (t) - t graph as shown in Figure 2.33.

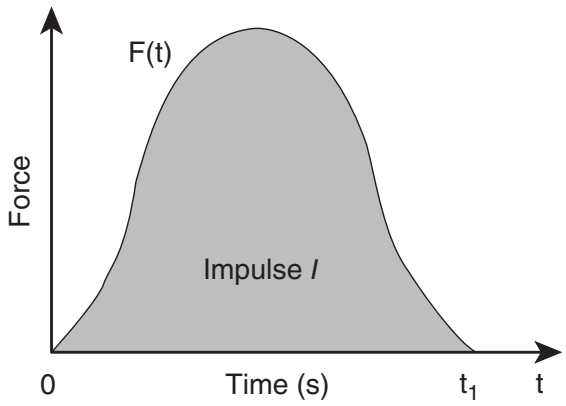


Figure 2.33 Impulse of a force

Average velocity

Consider a particle located initially at point P having position vector $\vec{r}_1$. In a time interval Δt the particle is moved to the point Q having position vector $\vec{r}_2$. The displacement vector is $\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$. This is shown in Figure 2.34.